Question 1

①
$$\begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 0 \end{bmatrix}$$

$$\boxed{\begin{aligned} x_1 &= x \\ x_2 &= y \\ x_3 &= z \end{aligned}}$$

Determinant

$$= \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad 2\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} - 1\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - 1\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$2(0-1) - 1(1-1) - 1[1-0]$$

$$-2 \quad 0 \quad -1$$

$$D = -3$$

$$D_x = \begin{vmatrix} 0 & 1 & -1 \\ 4 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} \quad 0\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} - 1\begin{bmatrix} 4 & 1 \\ 0 & 1 \end{bmatrix} - 1\begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$$

$$0[0-1] - 1(4-0) - 1(4-0)$$

$$-4 \; -4$$

$$D_x = -8$$

$$D_y = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 4 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad 2\begin{bmatrix} 4 & 1 \\ 0 & 1 \end{bmatrix} - 0\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - 1\begin{bmatrix} 1 & 4 \\ 1 & 0 \end{bmatrix}$$

$$2(4-0) \quad 0 \qquad -1\,(0-4]$$

$$8 \; +4$$

$$D_y = 12$$

$$D_z = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 4 \\ 1 & 1 & 0 \end{bmatrix} \quad 2\begin{bmatrix} 0 & 4 \\ 1 & 0 \end{bmatrix} \; -1\begin{bmatrix} 1 & 4 \\ 1 & 0 \end{bmatrix} \; 0\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$2(0-4) \quad -1(0-4) \qquad 0$$

$$-8 \qquad +4$$

$$= -4$$

to find $x_1 = x = \frac{D_x}{D} = \frac{-8}{-3} = \frac{8}{3} = 2\frac{2}{3}$

$x_2 = y = \frac{D_y}{D} = \frac{12}{-3} = -4$

$x_3 = z = \frac{D_z}{D} = \frac{+4}{+3} = \frac{4}{3} = 1\frac{1}{3}$

therefore; $x_1 = \frac{8}{3}$

$x_2 = -4$

$x_3 = \frac{4}{3}$

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Question 2

$6x - y = 0$

$2x - 4y = 1$

$$\begin{bmatrix} 6 & -1 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$A \cdot X = C$

$A \cdot X = C$

$A^{-1} \cdot A \cdot X = A^{-1}C$

$X = A^{-1}C$

$\cancel{A}$

$X = A^{-1}C$

To find the inverse of $A^{-1} = \frac{1}{\text{determint}} \times \text{Adjont}$

$A^{-1} = \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$A^{-1} = \frac{1}{(6x-4)-(2x-1)} \begin{bmatrix} -4 & 1 \\ -2 & 6 \end{bmatrix}$

$A^{-1} = \frac{1}{-24+2} \begin{bmatrix} -4 & 1 \\ -2 & 6 \end{bmatrix}$

$A^{-1} = \frac{1}{-22} \begin{bmatrix} -4 & 1 \\ -2 & 6 \end{bmatrix}$

$A^{-1} = \begin{bmatrix} \frac{1}{-22} \times -4 & \frac{1}{-22} \times 1 \\ \frac{1}{-22} \times -2 & \frac{1}{-22} \times 6 \end{bmatrix} = \begin{bmatrix} \frac{2}{11} & \frac{1}{-22} \\ \frac{1}{11} & -\frac{3}{11} \end{bmatrix}$

$\cancel{X} = \begin{bmatrix} \frac{2}{11} & \frac{1}{-22} \\ \frac{1}{11} & -\frac{3}{11} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$X = \begin{bmatrix} (\frac{2}{11} \times 0) + (\frac{1}{-22} \times 1) \\ (\frac{1}{11} \times 0) + (-\frac{3}{11}(1)) \end{bmatrix}$ $X = \begin{bmatrix} 0 + \frac{1}{-22} \\ 0 + -\frac{3}{11} \end{bmatrix}$ $X = \begin{bmatrix} \frac{1}{-22} \\ -\frac{3}{11} \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$

$x = \frac{1}{-22}$ $y = \frac{-3}{11}$

Question 3a

3a

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 3 & 4 & 3 \\ 1 & -2 & -1 & 1 \end{array}\right] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Reducing to $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 3 & 4 & 3 \\ 1 & -2 & -1 & 1 \end{array}\right] = R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 3 & 4 & 3 \\ 0 & -3 & -2 & -1 \end{array}\right] = R_2 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & -3 & -2 & -1 \end{array}\right] = R_3 + 3R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 4 & -4 \end{array}\right] = R_3 \div 4 \qquad = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 1 & -1 \end{array}\right] = R_2 - 2R_3$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array}\right] = R_1 - R_3 \qquad = \left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array}\right] = R_1 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array}\right]$$

$$\therefore \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$x_1 = 2 \quad x_2 = 1 \quad x_3 = -1$

Question 3b

(3b) $x_1+x_2+x_3=2$

$2x_1+3x_2+4x_3=3$

$x_1-2x_2-x_3=1$

$$\begin{bmatrix}1&1&1\\2&3&4\\1&-2&-1\end{bmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}=\begin{bmatrix}2\\3\\1\end{bmatrix}$$

finding the Inverse.

Find Determinant.

$$Det = 1\begin{vmatrix}3&4\\-2&-1\end{vmatrix} - 1\begin{vmatrix}2&4\\1&-1\end{vmatrix}\ 1\begin{vmatrix}2&3\\1&-2\end{vmatrix}$$

$$= 1(-3+8)-1(-2-4)\ 1(-4-3)$$

$$= 5+6-\cancel{8}7$$

$$= 5+6-7$$

$$Det = 4$$

find ~~Adjoint~~

$$\begin{bmatrix}+\begin{vmatrix}3&4\\-2&-1\end{vmatrix} & -\begin{vmatrix}2&4\\1&-1\end{vmatrix} & +\begin{vmatrix}2&3\\1&-2\end{vmatrix}\\ -\begin{vmatrix}1&1\\-2&-1\end{vmatrix} & +\begin{vmatrix}1&1\\1&-1\end{vmatrix} & -\begin{vmatrix}1&1\\1&-2\end{vmatrix}\\ +\begin{vmatrix}1&1\\3&4\end{vmatrix} & -\begin{vmatrix}1&1\\2&4\end{vmatrix} & +\begin{vmatrix}1&1\\2&3\end{vmatrix}\end{bmatrix} = \begin{bmatrix}+5 & +6 & -7\\-1 & -2 & 3\\+1 & -2 & 1\end{bmatrix}$$

Adjoint.

$$inverse = \frac{1}{4}\left[\begin{pmatrix}5&6&-7\\-1&-2&3\\1&-2&1\end{pmatrix}\right] = \begin{bmatrix}5&-1&1\\6&-2&-2\\-7&3&1\end{bmatrix}$$

$$inverse = \frac{1}{4}\begin{bmatrix}5&-1&1\\6&-2&-2\\-7&3&1\end{bmatrix}$$

$$Inverse = \begin{bmatrix}\frac{5}{4}&-\frac{1}{4}&-\frac{1}{4}\\ \frac{6}{4}&\frac{-2}{4}&\frac{-2}{4}\\ \frac{-7}{4}&\frac{3}{4}&\frac{1}{4}\end{bmatrix}$$

$$\text{Therefore Inverse} = \begin{bmatrix}\frac{5}{4}&-\frac{1}{4}&-\frac{1}{4}\\ \frac{3}{2}&-\frac{1}{2}&-\frac{1}{2}\\ -\frac{7}{4}&\frac{3}{4}&\frac{1}{4}\end{bmatrix}$$